Explore Sinusoids Results

Copyright Barry Van Veen 2016

Contents

- Set up workspace
- Problem 1
- Problem 2
- Problem 3

Set up workspace

```
clear
close all
```

Problem 1

```
t = -1:0.001:1;

% Part a

x = 5*cos(2*pi*t);

figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part a, f = 1 Hz, \phi = 0 ')

% Part b

x = 5*cos(2*pi*t - pi/4);

figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part b, f = 1 Hz, \phi = -\pi/4 ')

% Part c

x = 5*cos(2*pi*2*t - pi/4);

figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part c, f = 2 Hz, \phi = -\pi/4 ')

% Part d
```

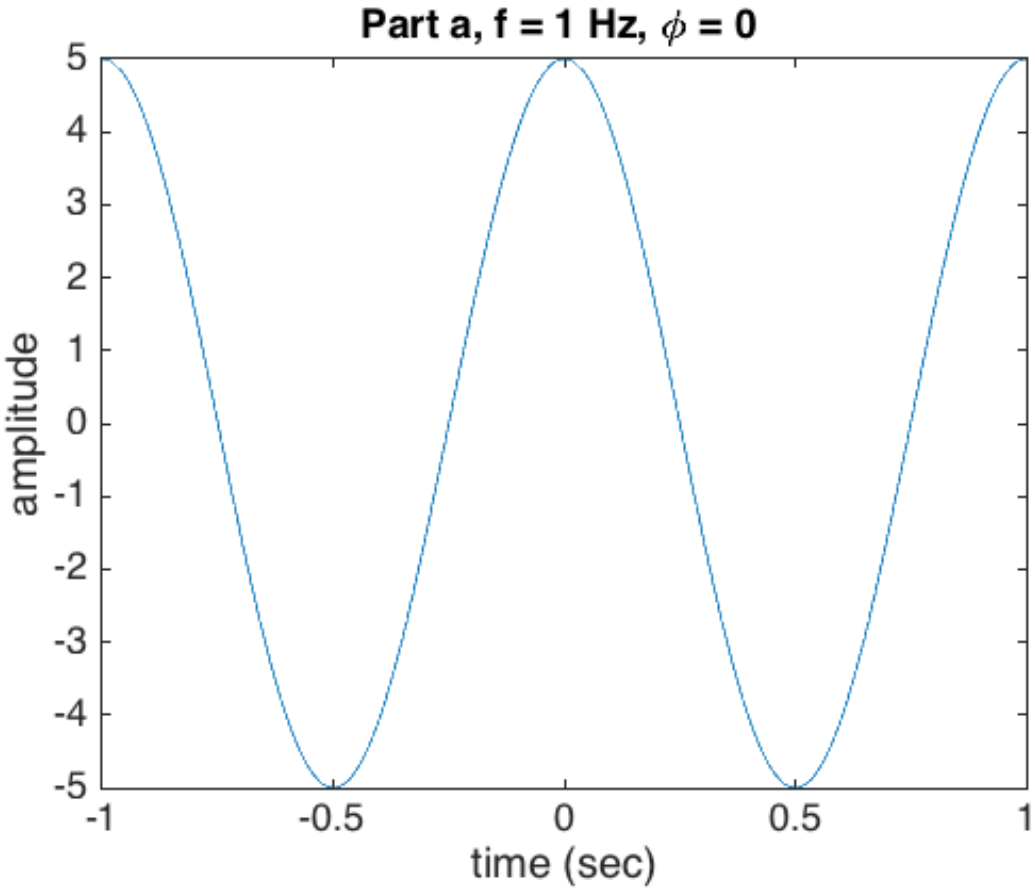
```
x = 5*cos(2*pi*5*t - pi/4);

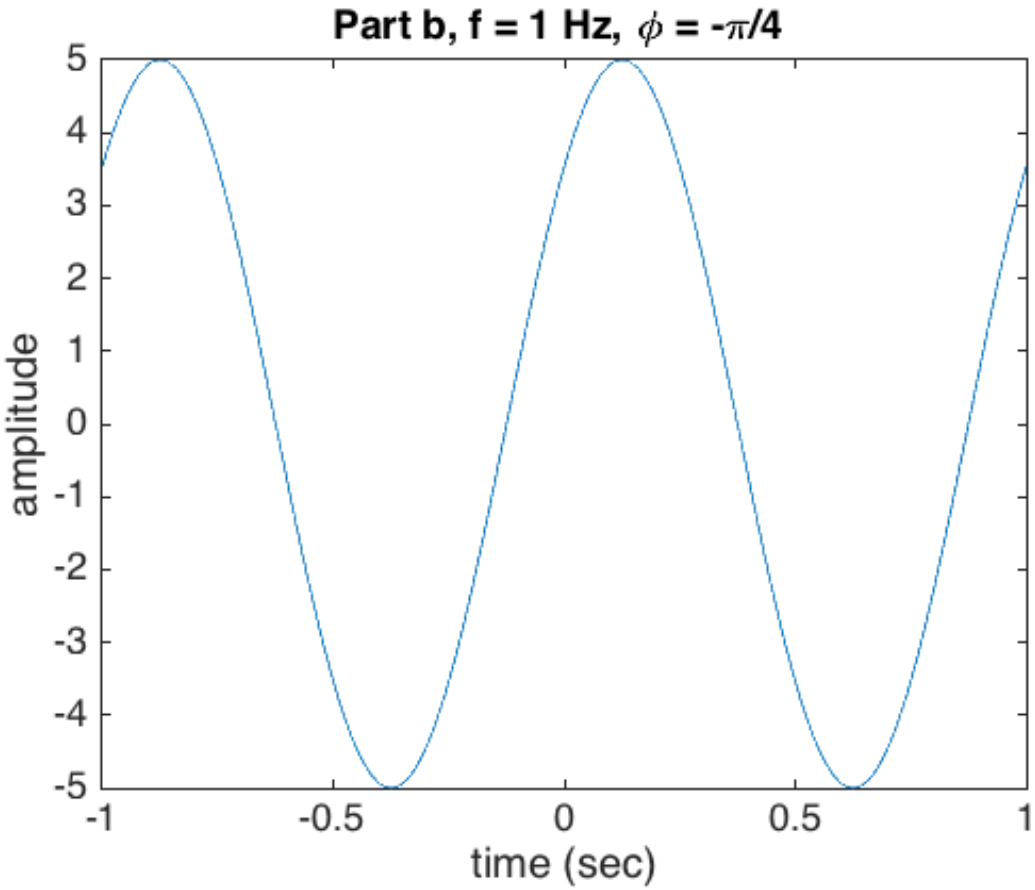
figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part d, f = 5 Hz, \phi = -\pi/4 ')

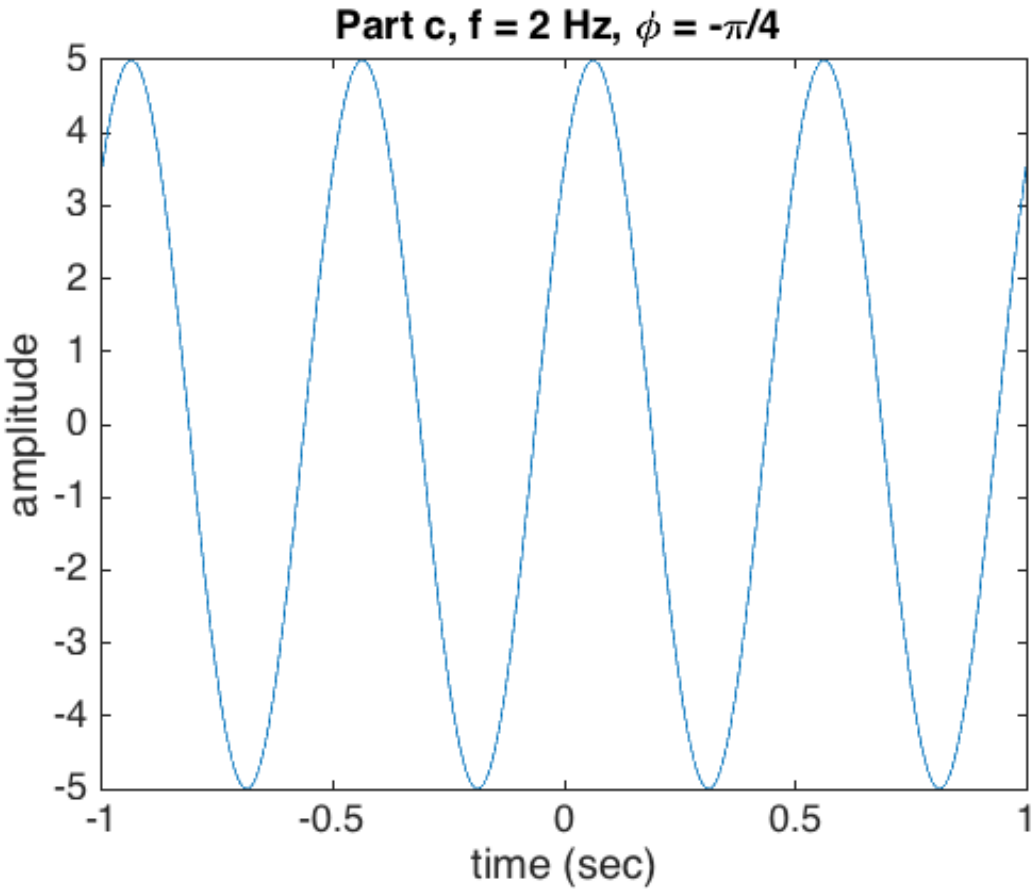
% Part e

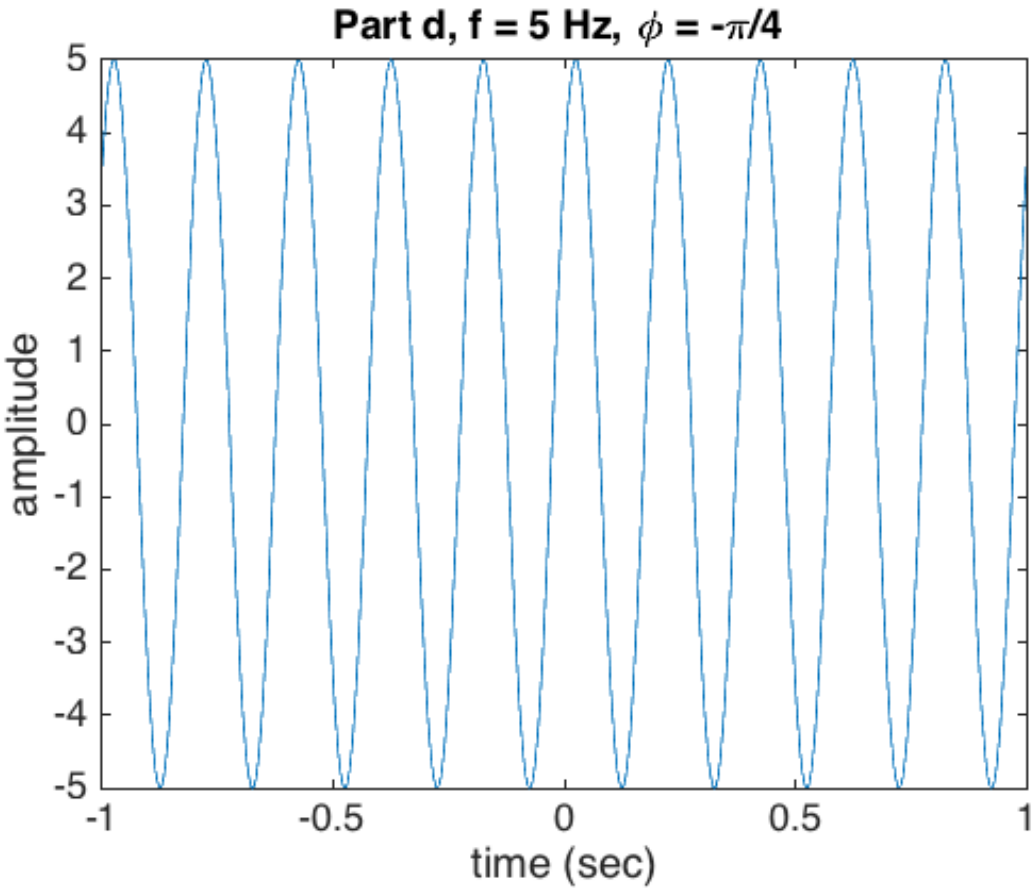
x = 5*cos(2*pi*10*t - pi/4);

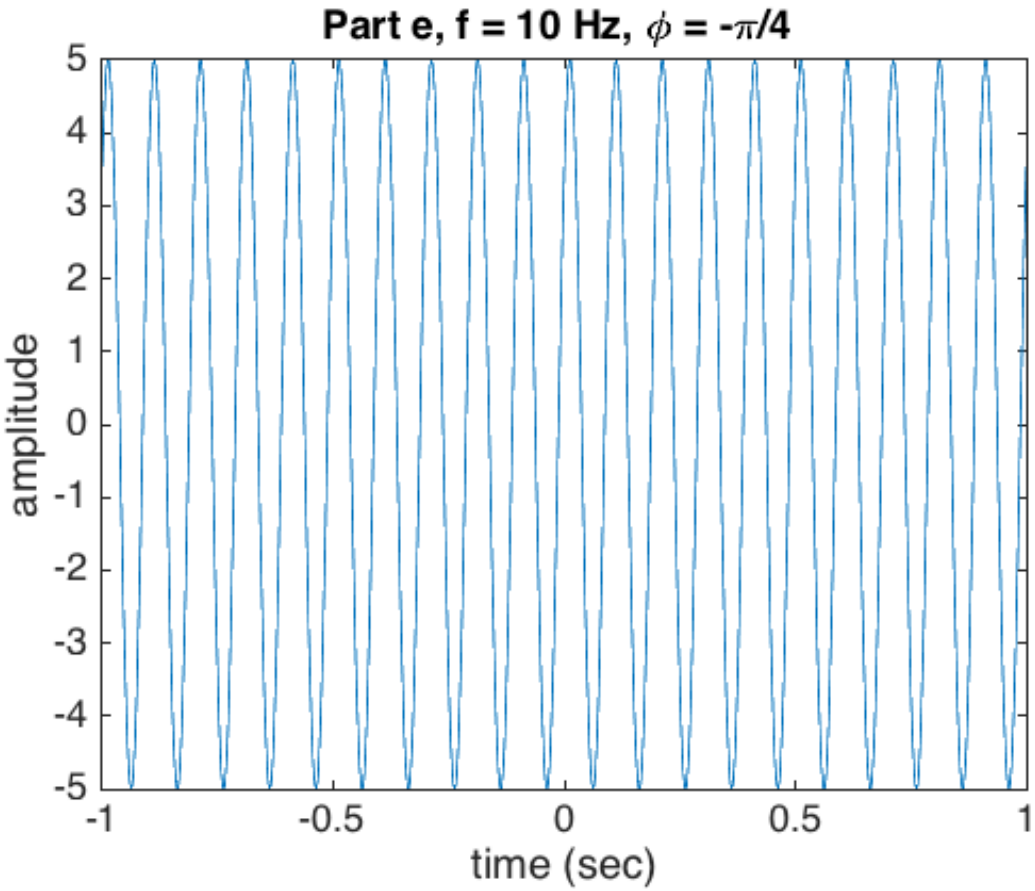
figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part e, f = 10 Hz, \phi = -\pi/4 ')
```




Part b, f = 1 Hz, ϕ = -π/4
amplitude
time (sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1
-0.5
0
0.5
1


Part c, f = 2 Hz, ϕ = -π/4
amplitude
time (sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1
-0.5
0
0.5
1


Part d, f = 5 Hz, ϕ = -π/4
amplitude
time (sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1
-0.5
0
0.5
1


Part e, f = 10 Hz, ϕ = -π/4
amplitude
time (sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1
-0.5
0
0.5
1

Problem 2

```
n = -10:90;

% Part a

x = 3*cos(pi*n/50 + pi/4);

figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
xlim([-10,90])
title('Part a, \Omega = \pi/50 rads, \phi = \pi/4 ')

% Part b

x = 3*cos(pi*n/5);

figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
title('Part b, \Omega = \pi/5 rads, \phi = 0 ')
xlim([-10,90])

% Part c

x = 3*cos(pi*n/2);

figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
title('Part c, \Omega = \pi/2 rads, \phi = 0 ')
xlim([-10,90])

% Part d

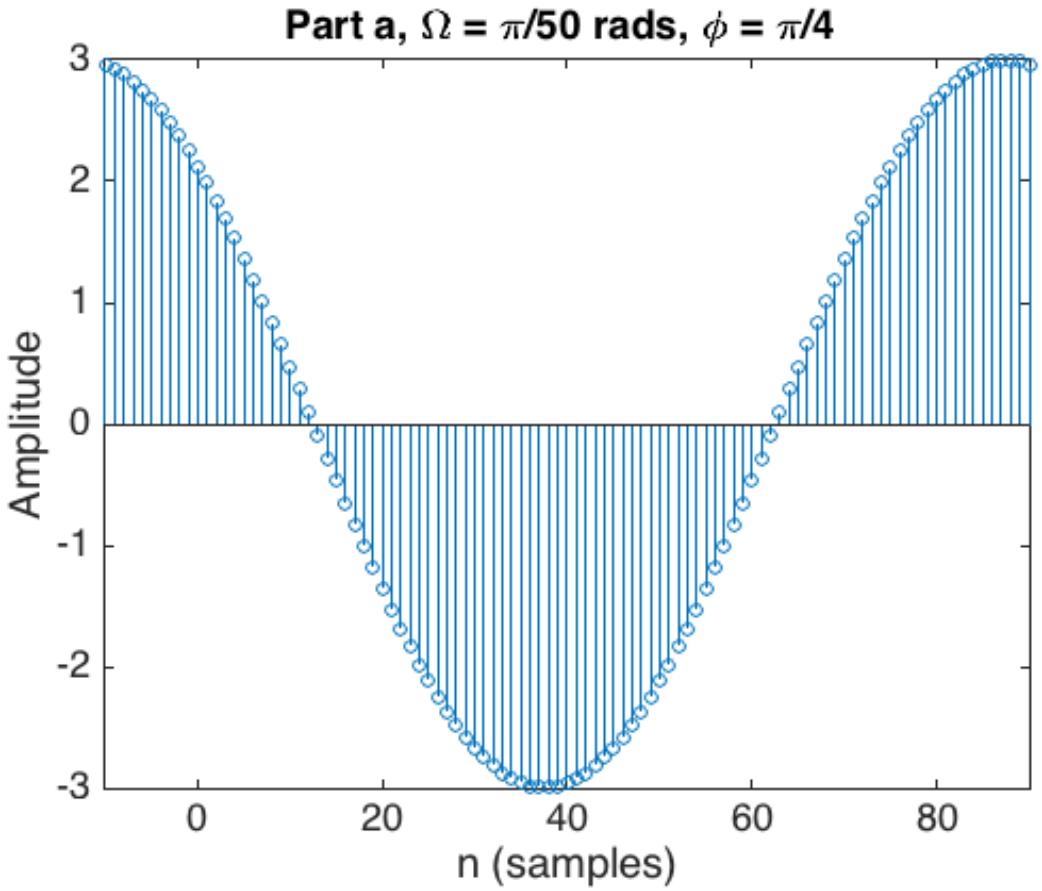
x = 3*cos(11*pi*n/5);

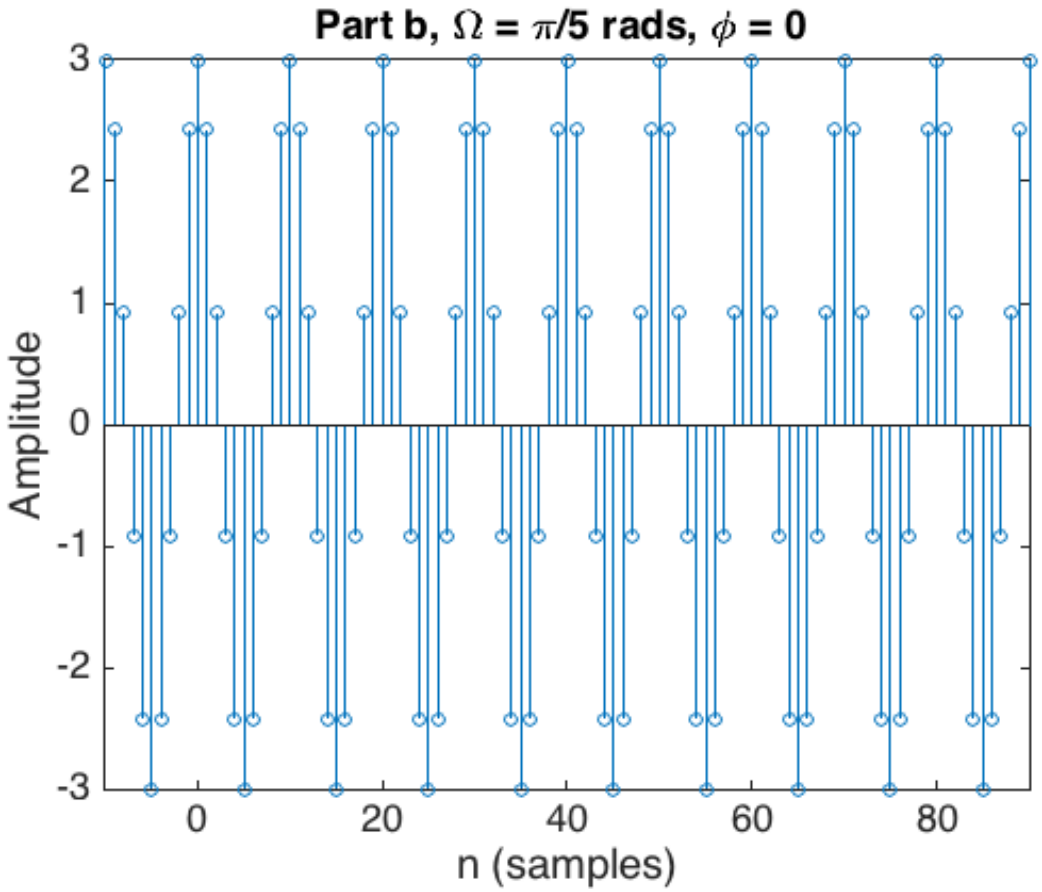
figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
title('Part d, \Omega = 11\pi/5 rads, \phi = 0 ')
xlim([-10,90])

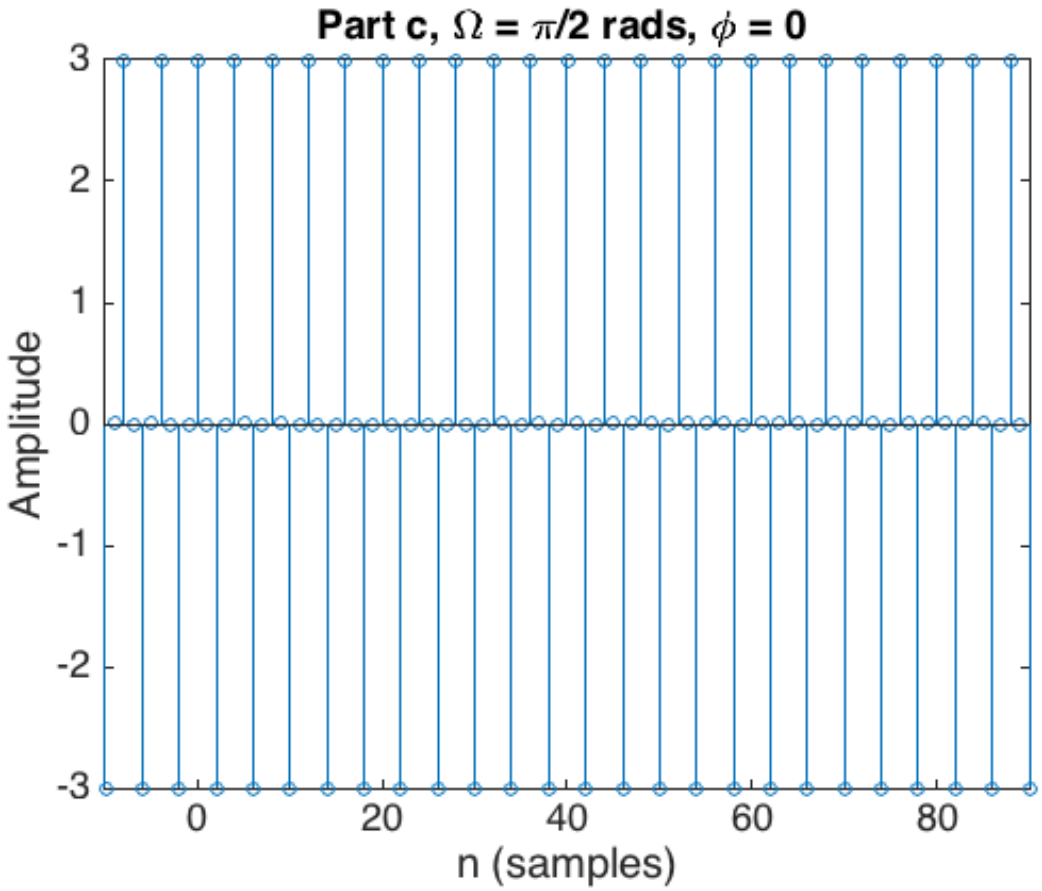
% Part e

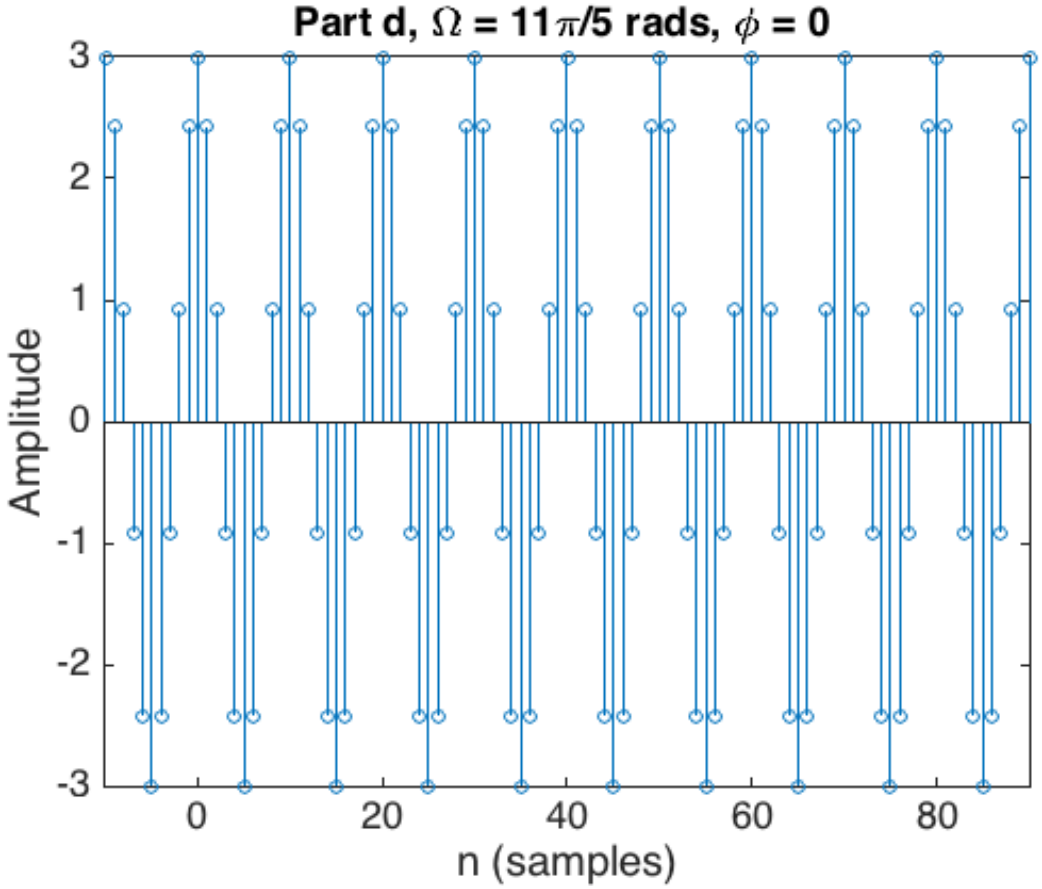
x = 3*cos(21*pi*n/5);

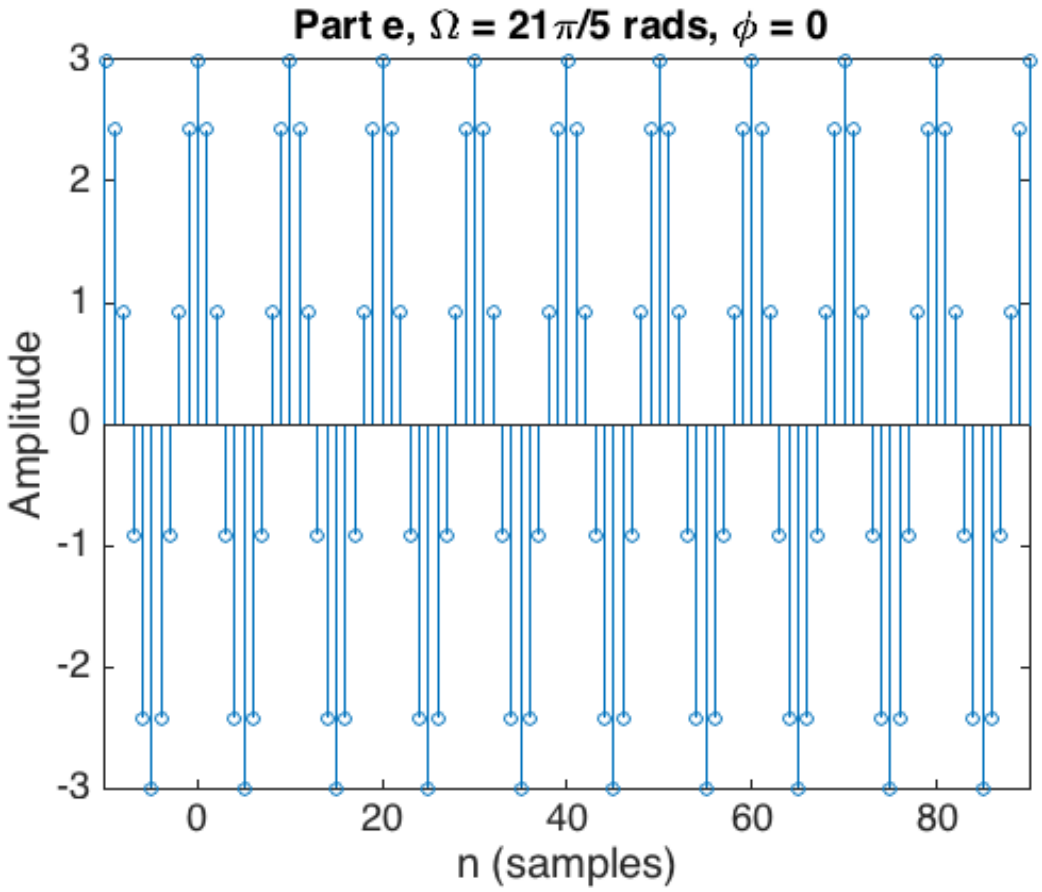
figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
title('Part e, \Omega = 21\pi/5 rads, \phi = 0 ')
xlim([-10,90])
```


Part a, Ω = π/50 rads, ϕ = π/4
Amplitude
n (samples)
3
2
1
0
-1
-2
-3
0
20
40
60
80


Part b, Ω = π/5 rads, ϕ = 0
Amplitude
n (samples)
3
2
1
0
-1
-2
-3
0
20
40
60
80


Part c, Ω = π/2 rads, ϕ = 0
Amplitude
n (samples)
3
2
1
0
-1
-2
-3
0
20
40
60
80


Part d, Ω = 11π/5 rads, ϕ = 0
Amplitude
n (samples)
3
2
1
0
-1
-2
-3
0
20
40
60
80



Problem 3

```
t = 0:0.001:1;

j = sqrt(-1);

c = exp(j*3*pi*t);
s = exp(j*6*pi*t);

% Part a

c(50:54)

% Part b

y = 2*c + s;

figure
subplot(211)
plot(t,real(y))
title('Real part of y ')
ylabel('Amplitude')

subplot(212)
plot(t,imag(y))
title('Imaginary part of y ')
ylabel('Amplitude')
xlabel('Time (sec)')
```

```
% Part c

z = c.*s;

figure
subplot(211)
plot(t,abs(z))
title('Magnitude of c.*s ')
ylabel('Amplitude')
ylim([0 1.1])

subplot(212)
plot(t,angle(z))
title('Phase of c.*s ')
ylabel('Radians')
xlabel('Time (sec)')
```

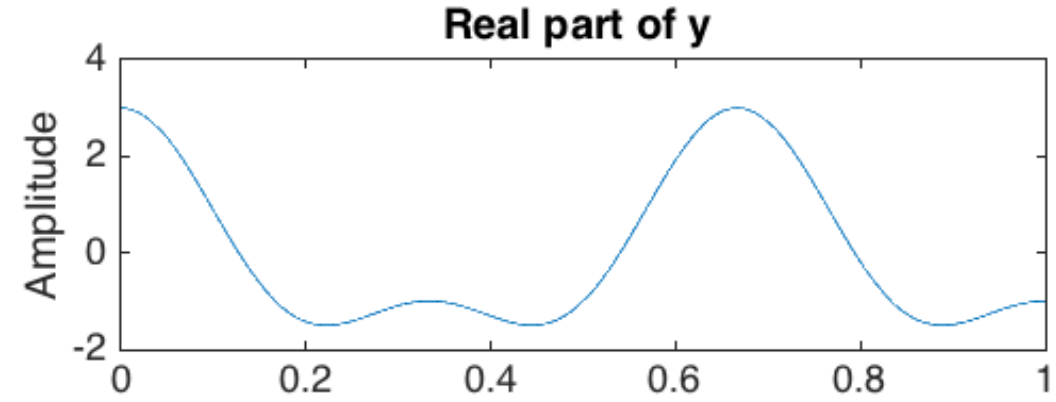
```
ans =

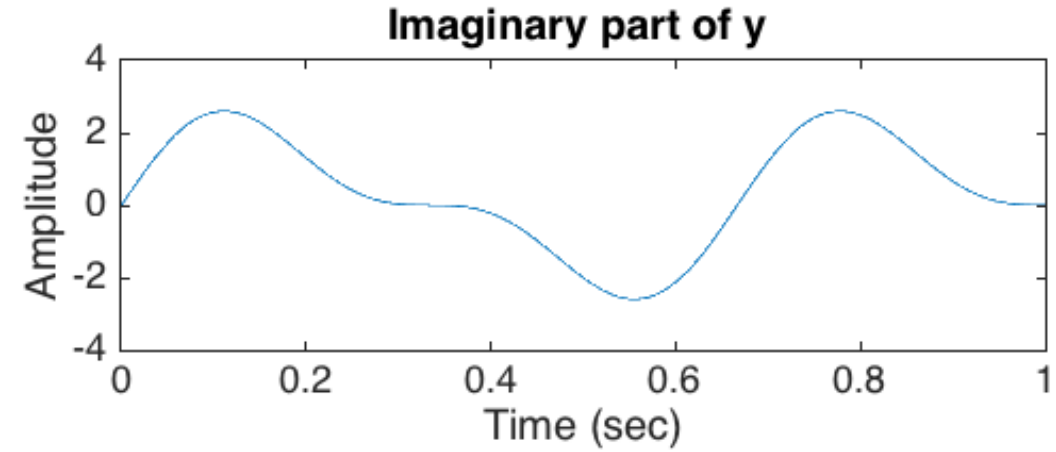
  Columns 1 through 4

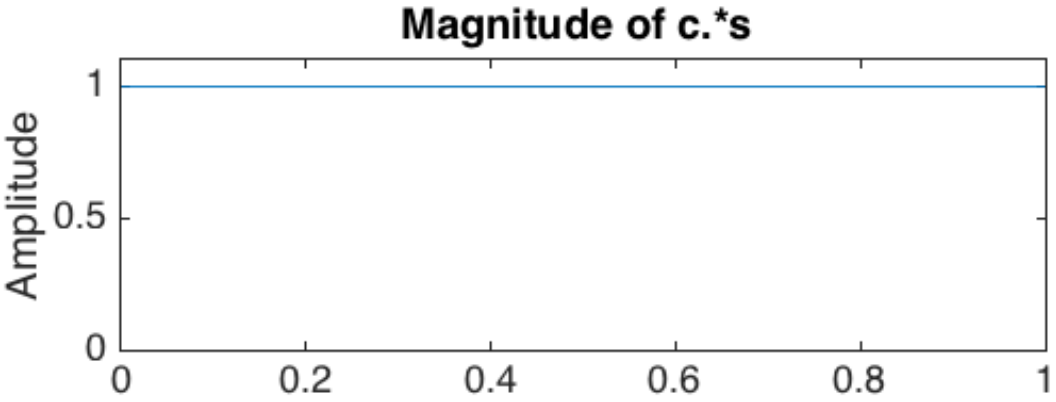
   0.8952 + 0.4456i   0.8910 + 0.4540i   0.8867 + 0.4624i   0.8823 + 0.4707i

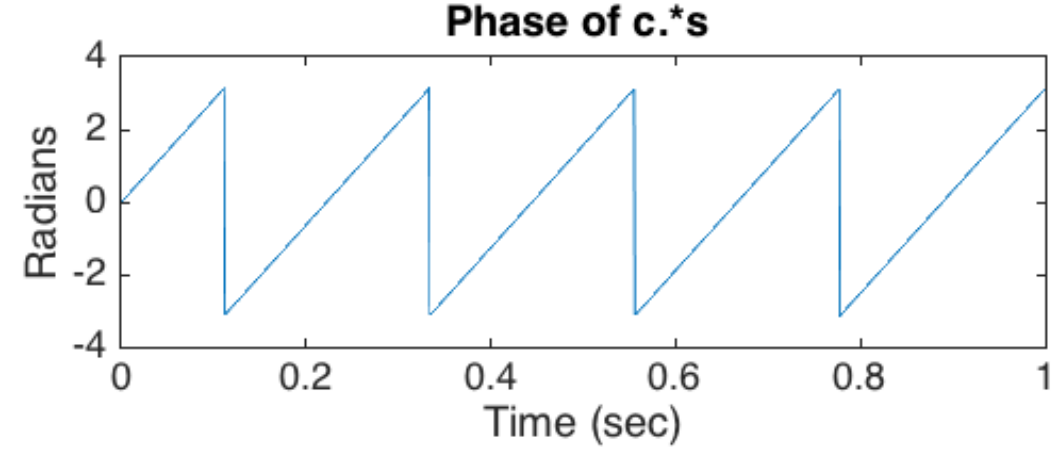
  Column 5

   0.8778 + 0.4790i
```


Real part of y
Amplitude
4
2
0
-2
0
0.2
0.4
0.6
0.8
1


Imaginary part of y
Amplitude
4
2
0
-2
-4
0
0.2
0.4
0.6
0.8
1
Time (sec)


Magnitude of c.*s
Amplitude
1
0.5
0
0
0.2
0.4
0.6
0.8
1


Phase of c.*s
Radians
4
2
0
-2
-4
0
0.2
0.4
0.6
0.8
1
Time (sec)

Published with MATLAB® R2015a